theory. Briefly put, the equations of string theory are so complicated that no one knows their exact form. Physicists have managed to write down only approximate versions of the equations. It is these approximate equations that differ significantly from one string theory to the next. And it is these approximate equations, within the context of any one of the five string theories, that give rise to an abundance of solutions, a cornucopia of unwanted universes.

Since 1995 (the start of the second superstring revolution), there has been a growing body of evidence that the exact equations, whose precise form is still beyond our reach, may resolve these problems, thereby helping to give string theory the stamp of inevitability. In fact, it has already been established to the satisfaction of most string theorists that, when the exact equations are understood, they will show that all five string theories are actually intimately related. Like the appendages on a starfish, they are all part of one connected entity whose detailed properties are currently under intense investigation. Rather than having five distinct string theories, physicists are now convinced that there is one theory that sews all five into a unique theoretical framework, And like the clarity that emerges when hitherto hidden relationships are revealed, this union is providing a powerful new vantage point for understanding the universe according to string theory.

To explain these insights we must engage some of the most difficult, cutting-edge developments in string theory. We must understand the nature of the approximations used in studying string theory and their inherent limitations. We must gain some familiarity with the clever techniques—collectively called dualities—that physicists have invoked to circumvent some of these approximations. And then we must follow the subtle reasoning that makes use of these techniques to find the remarkable insights alluded to above. But don’t worry. The really hard work has already been done by string theorists and we will content ourselves here with explaining their results.

Nevertheless, as there are many seemingly separate pieces that we must develop and assemble, in this chapter it is especially easy to lose the forest for the trees. And so, if at any time in this chapter the discussion gets a little too involved and you feel compelled to rush on to black holes (Chapter 13) or cosmology (Chapter 14), take a quick